Metoprolol Tartrate

Mechanism of action

- Metoprolol is an immediate release beta blocker. It antagonises (blocks) beta-1 receptors in the heart, causing a decrease in heart rate, cardiac output and blood pressure.

Authorised practice levels

- Paramedics, ICPs and CCPs.

Indications

- Adults with moderate cardiovascular compromise as a result of atrial fibrillation, where the ventricular rate is predominantly greater than 130/minute.

Contraindications

- Known severe allergy.
- Hypotension. Metoprolol will further reduce the blood pressure.

Cautions

- First degree heart block. Metoprolol may cause bradycardia.
- Known sick sinus syndrome without an internal pacemaker in place. Metoprolol may cause bradycardia.
- Previous second or third degree heart block without an internal pacemaker in place. Metoprolol may cause worsening of heart block and bradycardia.
- Asthma or COPD. Metoprolol may cause bronchospasm and should usually be withheld if the patient regularly takes bronchodilators.
- Heart failure. Metoprolol will reduce cardiac output and may make heart failure worse.
- Pregnancy.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy. The likelihood of administration being required in a woman who is pregnant is so low that personnel must seek clinical advice prior to administration.

Metoprolol Tartrate

- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- 25 mg.

Administration

- Administer PO.

Common adverse effects

- None.

Usual onset of effect

- 10-20 minutes

Usual duration of effect

- 4-6 hours

Usual preparation

- 50 mg tablets.

Pharmacokinetics

- Metoprolol is absorbed within the gastrointestinal tract and metabolised in the liver.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- The blood pressure effect will be potentiated by other medicines that lower blood pressure. For example, GTN, antihypertensive medicines and amiodarone.

Metoprolol Tartrate

- The heart rate effects will be potentiated by other medicines that lower the heart rate. For example, amiodarone and centrally acting calcium channel blockers such as diltiazem.

Additional information

- Patients with chronic or paroxysmal atrial fibrillation may be prescribed a beta blocker. Personnel should still administer metoprolol as indicated, even if the patient has taken their own beta blocker.

More Information

Version history

16 June 2026 1.1.0.1

Update: Indications and dosage updated.

11 September 2023 1.0.4

New: Guideline released.